

STAR MELEE — Complete Development Plan

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Purpose: Comprehensive build specification for the iOS / iPadOS / macOS game **Star Melee**, optimized for use with Claude Code.

Inspiration: *Star Control* and *Star Control II* (Super Melee combat mode), with original IP, original ships, original assets, and significant gameplay extensions.

How to Use This Document with Claude Code

This document is structured so that Claude Code can:

1. Read the whole document first to understand scope.
2. Implement features section-by-section in the order presented.
3. Stop and prompt the user at >>> **DECISION POINT** <<< markers.
4. Fall back to reasonable defaults (Star Control II-inspired behavior) whenever the spec is silent.

Working agreement with Claude Code:

- Use Git from day one. Commit after every working feature.
- Build minimum-viable versions first, then iterate.
- Run on a physical device early and often.
- Reference [star-melee-mockup.html](#) for visual/mechanical intent.

1. Project Vision & Scope

Game Identity

- **Name:** Star Melee
- **Tagline:** Strategic ship-to-ship arena combat across the stars.
- **Genre:** 2D real-time space combat / arena fighter
- **Inspiration:** *Star Control II* Super Melee combat depth × *Mortal Kombat 1* match structure and presentation.

Monetization Model

Freemium:

- **Free tier:** Core game with a starter roster of ships (recommend 6–8).
- **Paid expansion packs:** Additional ships (3–6 ships per pack).
- **Weapon enhancements:** Upgrade modules (better fire rate, larger shield capacity, enhanced special, etc.) for individual ships.
- **No ads in v1.0.** Re-evaluate after launch.

DECISION POINT: Final free vs. paid ship split. Recommendation: 6 free starter ships, 2 paid packs of 3 ships each at launch (\$1.99 each or \$4.99 bundle).

Platforms

- **Primary:** iOS (iPhone) and iPadOS — touch controls
- **Secondary:** macOS — keyboard/mouse controls (via Mac Catalyst or native AppKit/SwiftUI)
- **Future:** Apple TV with controller support (Phase 5+)

Target Performance

- 60 FPS on iPhone 12 and newer, iPad Air (4th gen) and newer, Apple Silicon Macs.
- App size under 250 MB for v1.0.

2. Technology Stack

Recommended Stack: Swift + SpriteKit + SwiftUI

- **SpriteKit** for the 2D arena combat (rendering, physics primitives, particle effects).
- **SwiftUI** for menus, HUD overlays, settings, Ship Compendium screens.
- **SceneKit** for the 3D ship rotation views in the Ship Compendium.
- **Core Haptics** for per-weapon and per-event haptic feedback.
- **AVFoundation** for music; **AVAudioEngine** for spatial sound effects.
- **GameKit** for Game Center leaderboards (optional in v1.0, recommended for v1.1).
- **StoreKit 2** for in-app purchases (ship packs, weapon enhancements).
- **Mac Catalyst** to ship the macOS version from the same codebase.

Why This Stack

- Native, no engine licensing fees, no royalties.
- Excellent Claude Code support (Swift + Apple frameworks are well-documented and well-understood by Claude).

- Maximum performance and battery life on Apple devices.
- The user already has iOS app development experience.

DECISION POINT: Confirm Swift + SpriteKit before scaffolding. If Unity is preferred (e.g., for easier 3D ship models and cross-platform expansion), flag now — switching mid-build is expensive.

3. Legal & IP Guidelines

- **No use of the names** "Star Control," "Ur-Quan," "Spathi," "Mycon," or any other trademarked or copyrighted IP from the Star Control series.
- **No copying of original ship designs**, alien races, dialog, music, or art assets.
- **All Star Melee ships, names, lore, art, music, and sound effects must be 100% original.**
- **Reference repositories (UQM, TW-Light) are for study only** — do not copy substantial code blocks. Reimplement mechanics from scratch in Swift.
- **App Store description:** Use phrasing like "inspired by classic 90s space combat." Avoid naming the original game.

Reference Repositories (study only)

- **The Ur-Quan Masters MegaMod** — <https://github.com/JHGuitarFreak/UQM-MegaMod> (study ship physics, weapon systems, AI)
- **TW-Light** — <https://github.com/Yurand/tw-light> (study melee-focused combat loop)

4. Core Gameplay Loop

Match Structure (Mortal Kombat-style)

1. **Main Menu** → Play, Compendium, Settings, Leaderboard
2. **Character Select Screen** → Player picks their ship; opponent is "Random" by default (or player selects)
3. **Pre-Match Countdown** — 10 second practice period.
 - **Player can:** move (D-pad/X/Y), fire **A (primary)** and **B (secondary)** weapons.
 - **AI can:** move and fire at *reduced rate* (~60% primary frequency, ~20% secondary frequency) so practice feels real but is survivable.
 - **Locked during countdown for both ships:** C (special weapon), Z (speed boost), A+B (Transporter Beam), B+C (Cloak), A+C (Self-Destruct).
 - **Locked during countdown:** power-up spawning.
 - **Visual:** the big countdown digit must be rendered at ~35% opacity with an outline so the arena stays visible behind it. A "PRACTICE" banner with hint text appears above the ships.
4. **Match Begins** — 2-minute timer starts. Full combat.

5. **Match Ends** when either ship is destroyed, OR timer expires (ship with higher remaining health wins).
6. **Next Match** begins 3 seconds after the previous one ends. Health, shield, battery all reset to 100%.
7. **Series Winner** — first to 2 wins out of 3 matches.
8. **Victory Screen** — Mortal Kombat-style "WINNER" announcement. If a Quantum Torpedo from a Transporter Beam delivered the killing blow, display "**FATALITY**" below the winner banner.
9. **Post-Match** — Replay, change ship, return to main menu.

Match Timer Behavior

- Countdown displayed at top center: 2:00 → 0:00.
- At 0:00, ship with higher remaining health % wins.
- If health tied: ship with higher shield % wins; if still tied, ship with higher battery % wins; if still tied, declare a Draw and replay match.
- **Timer Extension Power-Up**: +30 seconds when collected mid-match.

5. Ship System

Ship Performance Profile

Every ship has these stats, balanced across ships so each has a viable path to victory:

Stat	Description	Typical Range
Max Health	Life bar capacity	60–120
Heal Rate	Health regen per second	+0.5% to +3%
Max Shield	Shield bar capacity	0 (no shield) to 100
Shield Up/Down Time	Seconds to raise/lower shields	0.1s to 3.0s
Max Battery	Energy reserve	60–120
Battery Regen	+1% per 0.5s base, ship-modifier 0.5×–1.5×	
Mass	Affects gravity influence	0.6–1.8
Acceleration	Thrust force	0.05–0.20
Max Speed	Velocity cap	4–9 units/frame
Turn Rate	Radians per frame	0.03–0.09
Hitbox Size	Physical profile	Small (10px), Medium (14px), Large (20px)
Primary Weapon Fire Rate	Frames between shots	6–24
Secondary Weapon Fire Rate	Frames between shots	18–60
Special Weapon Cooldown	Seconds	8–25
Speed Boost Cooldown	Seconds	6–15

Stat	Description	Typical Range
Transporter Beam Cooldown	Seconds	20–40

Ship Roster (Initial v1.0 — 12 ships, 6 free + 6 in expansion packs)

ALLIANCE FACTION (defenders) — Free tier in v1.0

1. **AEGIS CRUISER** — Balanced workhorse. Medium speed/health/shield. Primary: rapid laser. Secondary: guided missile. Special: **Inertia Dampeners** (nullify planet gravity for 6 seconds).
2. **SOLAR WING** — Fast scout. High speed, low health, no shield. Primary: pulse cannon. Secondary: flak burst. Special: **Super Speed Burst** (3 seconds of 3× speed).
3. **TITAN BULWARK** — Heavy tank. Huge health, slow, strong shield. Primary: dual cannon. Secondary: cluster bomb. Special: **Invulnerability Shield** (4 seconds of immunity).
4. **PRISM HUNTER** — Mid-weight ranger. Primary: prism spread (3 beams). Secondary: focused beam. Special: **EM Blast** (disrupts opponent's engine + weapons for 3s).
5. **NOVA LANCER** — Glass cannon. Fast, fragile, no shield. Primary: lance beam. Secondary: heavy lance. Special: **Homing Missile** swarm (4 missiles).
6. **HALO SENTINEL** — Defensive support. Medium speed, very strong shield, slow shield up/down. Primary: short pulse. Secondary: omni-burst (8-way). Special: **Mimic** (copy opponent's ship for 10s).

DOMINION FACTION (raiders) — Expansion Pack #1 ("Dominion Vanguard") in v1.0

7. **VOID REAPER** — Heavy bruiser. High health, slow, medium shield. Primary: plasma cannon. Secondary: gravity well bomb. Special: **Cloaking Device** (8 seconds invisible to AI, translucent to player).
8. **SCARAB STRIKER** — Fast attacker. Primary: swarm cannon. Secondary: rapid darts. Special: **Super Speed** (5 seconds of 2.5× speed + free turning).
9. **OBSIDIAN MAW** — Massive ship, largest hitbox. Devastating offense, slow. Primary: shard volley. Secondary: scatter mines. Special: **Self-Destruct** (4-second timer, devastating blast in radius).
10. **WRAITH PHANTOM** — Stealth class. Primary: weak beam, fast fire. Secondary: phase shot. Special: **Cloaking Device** + Phase Shift (12s cloak).
11. **BONE SPEAR** — Fast skirmisher. Primary: bone shards. Secondary: piercing spear. Special: **Mimic + Speed** combo.
12. **CRIMSON TYRANT** — Flagship. Premium ship. Strong everywhere. Primary: twin cannon. Secondary: heavy torpedo. Special: **Transporter Beam** (built-in advantage; faster transporter cooldown than other ships).

DECISION POINT: Confirm the free / paid ship distribution. Also confirm Crimson Tyrant being a "premium" ship is acceptable (it is the highest-tier ship — may want to balance vs. perception of pay-to-win).

Universal Ship Capabilities

Every ship has:

- **Primary weapon** (A button) — high fire rate, low per-shot damage
- **Secondary weapon** (B button) — lower fire rate, higher damage per shot

- **Special weapon** (C button) — unique ability, long cooldown
- **Transporter Beam** (A+B combo) — universal capability if ship has Transporter Beam installed
- **Cloaking Device** (B+C combo) — only ships with this ability
- **Self-Destruct** (A+C combo) — universal, but range/damage varies
- **Speed Boost** (Z button) — universal, cooldown varies

DECISION POINT: Should the Transporter Beam be universal, or only available to ships that "install" it via the weapon enhancement IAP? Spec is ambiguous. Recommendation: every ship can use Transporter Beam, but cooldowns and battery cost vary widely.

6. Weapon System

Weapon Categories

Primary Weapons (fast fire, low damage)

- **Laser Cannon** — straight beam, fast
- **Pulse Cannon** — rapid energy bolts
- **Plasma Burst** — slow but powerful
- **Prism Spread** — 3-shot spread
- **Lance Beam** — focused fast beam
- **Twin Cannon** — paired shots
- **Bone Shards** — small fragments
- **Swarm Cannon** — rapid scatter

Secondary Weapons (slow fire, high damage)

- **Guided Missile** — homes toward opponent
- **Cluster Bomb** — explodes on impact, area damage
- **Heavy Torpedo** — slow, devastating
- **Focused Beam** — slow charge, high damage
- **Flak Burst** — wide spread
- **Phase Shot** — passes through shields once
- **Piercing Spear** — penetrates and damages multiple times
- **Scatter Mines** — drop in arena, persist until destroyed

Special Weapons (long cooldown, unique mechanics)

- **Homing Missiles** — swarm of 4 self-guiding missiles
- **EM Blast** — disrupts opponent's engine + weapons for 3s
- **Cloaking Device** — ship invisible to AI, translucent to player

- **Super Speed Burst** — 3× speed for 3 seconds
- **Inertia Dampeners** — immune to planet gravity for 6 seconds
- **Mimic** — copy opponent's ship and weapons for 10 seconds
- **Self-Destruct** — devastating blast, kills user and damages opponent
- **Invulnerability Shield** — immune to all damage for 4 seconds

The Ultimate Weapon: TRANSPORTER BEAM + QUANTUM TORPEDO

Activation: A + B button combo.

Requirements:

- Ship must have shields lowered.
- Lowering shields takes the ship's specific Shield Down time (0.1s to 3.0s).
- Must have a Quantum Torpedo in inventory (collected as a power-up, or starting ammo varies by ship).
- High battery cost.

Mechanic:

1. Player engages Transporter Beam → ship animation + Star Trek-style transporter effect + sound.
2. If opponent ship's shields are DOWN → Quantum Torpedo is transported onto the opponent's bridge.
3. 10-second countdown timer appears above the opponent's ship (visible to both ships).
4. **Defense Option 1:** If opponent has a Transporter Beam installed and battery to use it, they can transport the torpedo into space behind their ship (it detonates harmlessly behind them).
5. **Defense Option 2:** If opponent is within ~25% of arena range of the firing ship, they can transport the torpedo BACK to the firing ship. Strategic: the firing ship must MOVE OUT OF RANGE quickly after firing.
6. **Detonation:** When the 10-second timer hits zero, the Quantum Torpedo explodes:
 - Catastrophic damage to ship it is currently inside.
 - Triggers a **QUANTUM SINGULARITY EVENT** in the arena.

QUANTUM SINGULARITY EVENT

- After torpedo detonates, the arena warps. Visual effect: spacetime ripples, color shift, gravitational lensing.
- **Physical obstructions appear** — debris, broken hull fragments, micro-singularities — that damage BOTH ships on contact.
- These obstructions persist until the end of the current match.
- Visual: pulsing dark/light effect, semi-transparent debris with glowing edges.

Damage Scaling Formula

Weapons damage scales by:

$$\text{final_damage} = \text{base_damage} \times (1 + \text{weapon_weight} \times \text{ship_offensive_modifier}) \times (1 - \text{target_shield_modifier} \times \text{target_shield_strength}) \times (1 - \text{target_armor_modifier})$$

Where:

- **weapon_weight**: Slow heavy weapons (torpedoes, bombs) weight 1.5–2.5. Fast weapons (lasers, pulses) weight 0.4–0.8.

- `ship_offensive_modifier`: Per-ship balance number.
- `target_shield_strength`: 0 to 1, scales with current shield %.

DECISION POINT: Final damage formula and ship balance values. Claude Code should build a JSON config file (`Ships.json` and `Weapons.json`) so balancing can iterate without code changes. Recommend that we tune through playtesting once Phase 2 is complete.

7. Shield, Battery, Health Systems

Health (Life Bar)

- Range: 0% (destroyed) to 100% (full).
- Color-coded: green (100–70%), yellow (70–35%), red (35–0%).
- Mortal Kombat 1 visual style: ship name above bar, bar drains from inside toward edge.
- Each ship has a **Self-Heal Rate** (e.g., +0.5% to +3% per second).
- Self-heal pauses for 2 seconds after taking damage.

Shield

- Range: 0% to 100%.
- Located visually directly under the Life Bar.
- Some ships have no shield (Solar Wing, Nova Lancer, Wraith Phantom in some configs).
- Shield absorbs damage before health.
- **Shield Up/Down time** varies (0.1s to 3.0s) — strategic for transporter beam usage.
- Shield slowly recharges when not under damage: +5% per second after 3 seconds of no damage.
- Shield must be DOWN to use the Transporter Beam.

Battery

- Range: 0% to 100%.
- Located visually directly under the Shield Bar.
- Drained by:
 - Speed Boost (15% per use)
 - Transporter Beam (40% per use)
 - Cloaking Device (5% per second while cloaked)
 - Special Weapon (varies by ship: 20%–60%)
 - Inertia Dampeners (10% per second while active)
- Regenerates at +1% per 0.5 seconds (ship-specific multiplier 0.5× to 1.5×).
- Resets to 100% at start of every match.
- Restored by power-ups (10%, 50%, 100%).

8. Power-Ups

Power-Up Spawn Logic

- **Spawn frequency:** every 15–30 seconds during a match (random).
- **Adaptive boost:** If a ship's health is below 25%, increase power-up spawn rate and prioritize spawning a power-up near that ship.
- **Visual:** Floating glowing icons in the arena. Approach to collect.
- **Despawn:** Power-up disappears after 12 seconds if uncollected.

Power-Up Types

1. **Life Restore** — three tiers: +10%, +50%, +100% (full).
2. **Battery Restore** — three tiers: +10%, +50%, +100% (full).
3. **Shield Restore** — +50% shield instantly.
4. **Quantum Torpedo Ammo** — adds 1 torpedo to inventory.
5. **Speed Boost Charge** — instantly resets Speed Boost cooldown.
6. **Special Weapon Reset** — instantly resets Special Weapon cooldown.
7. **Timer Extension** — adds 30 seconds to match timer.
8. **Damage Multiplier** — 2× outgoing damage for 8 seconds.
9. **Shield Regen Boost** — 3× shield regen rate for 10 seconds.
10. **Repair Drone** — auto-heals +1% per second for 15 seconds (stacks with ship heal rate).

Power-Up Indicators (HUD)

- Active power-up effects appear as **colored translucent icons** below the player's Battery bar.
- Each icon shows a countdown ring around it.
- Maximum 4 active power-ups visible at once.

9. Touch Controls (iOS / iPadOS)

Layout

Following the **Sega Genesis / 8BitDo M30** controller layout:

Bottom-left: Directional crosspad (D-pad)

- Up: rotate ship forward direction (or use X for thrust + D-pad for steering)

- Left: turn left
- Right: turn right
- Down: secondary maneuver / reverse

Bottom-right: Two rows of three buttons each (semi-transparent):

- **Top row:** X / Y / Z
- **Bottom row:** A / B / C

Button Map

Button	Function	Hold Behavior
A	Fire primary weapon	Hold = auto-fire at ship's primary fire rate
B	Fire secondary weapon	Hold = auto-fire at ship's secondary fire rate
C	Special weapon	Tap once (cooldown enforced)
X	Accelerate (thrust forward)	Hold to maintain thrust
Y	Brake (active deceleration)	Hold to brake
Z	Speed Boost	Tap (cooldown enforced)
A + B	Engage Transporter Beam → Quantum Torpedo	—
B + C	Engage Cloaking Device	—
A + C	Begin Self-Destruct timer	—

Gesture Controls (additional)

- **Tap on screen:** Target weapon (for targeted weapons like homing missile)
- **Tap and swipe on own ship:** Move ship in swipe direction
- **Tap and hold on own ship + drag:** Continuous movement following finger
- **Double-tap anywhere:** Pause game

Pause Menu

- **Resume** — return to match
- **Restart Match** — restart current match (counts as a loss)
- **Ship Compendium** — view ship details
- **Settings**
- **Quit to Main Menu** (counts as a forfeit)

10. macOS Controls

Keyboard Map

Key	Function
W or ↑	Thrust (X equivalent)
S or ↓	Brake (Y equivalent)
A or ←	Turn left
D or →	Turn right
Space	Fire primary (A button)
F	Fire secondary (B button)
G	Special weapon (C button)
Shift	Speed boost (Z button)
T	Transporter Beam (A+B equivalent)
C	Cloaking Device (B+C equivalent)
K	Self-Destruct (A+C equivalent)
P or Esc	Pause
Tab	Target lock (cycle targets)

Mouse

- **Click on enemy ship:** Lock target for targeted weapons.
- **Click and drag own ship:** Direct movement (mirrors touch behavior).

11. Ship Compendium

Purpose

Detailed view of each ship's full performance profile. Accessible from main menu and pause menu.

Layout per Ship Page

- **3D Ship Model** — rotatable 360° with pinch-to-zoom (iOS) or click-drag (macOS).
- **Ship Name** — large display font.
- **Faction Badge**
- **Performance Stats** — all stats from Section 5 (Ship System) shown as bars.
- **Weapon Loadout** — primary, secondary, special with descriptions.
- **Strengths** — bullet list of pros.
- **Weaknesses** — bullet list of cons.

- **Recommended Play Style** — short paragraph.
- **Win/Loss Stats** — player's win-loss record with this ship.
- **Unlock Status** — Free / Owned / Locked (with link to IAP screen if locked).

Expandability

The Compendium must support adding new ships via:

- IAP unlock of ship pack
- Update releases adding new ships
- New ship entries should slot in alphabetically within their faction.

DECISION POINT: For v1.0, can 3D ship models be **stylized low-poly geometric models** (built programmatically with SceneKit)? This is the recommended MVP approach. Higher-fidelity models can come in v1.1+.

12. Audio Design

Music

- **Style reference:** Hans Zimmer's "No Time for Caution" from *Interstellar* — driving, anxious, building, organ + percussion + strings.
- **Use:** Looping during matches. Variants for menu, victory, defeat, and Quantum Singularity Event (tense overlay).
- **For v1.0 MVP:** Generate placeholder track using royalty-free music libraries (free.tunes, Pixabay Music, Free Music Archive) styled to that aesthetic. Hire a composer post-launch.

Sound Effects (per-ship + per-event)

Each ship needs unique:

- **Engine hum** (looping while thrusting)
- **Primary weapon fire**
- **Secondary weapon fire**
- **Special weapon fire**
- **Shield raise / lower**
- **Damage taken** (varies by weapon type hitting it)
- **Destruction**

System sounds:

- **Power-up collect** (per power-up type)

- **Transporter Beam activation** — Star Trek-inspired shimmer (synthesize with high-frequency oscillation + envelope)
- **Quantum Torpedo countdown beep** (escalating pitch as timer nears zero)
- **Quantum Singularity Event** — deep boom + reality-warp tone
- **FATALITY** announcement
- **Match Start / Match End**

For v1.0 MVP

- Use Web Audio API-style synthesized sounds in Swift via AVAudioEngine.
- Generate procedural effects rather than recording.
- Acceptable quality for launch; replace with curated assets in v1.1.

13. Haptic Feedback (iOS Only)

Use **Core Haptics** for per-weapon and per-event feedback:

Event	Haptic Pattern
Primary weapon fire	Sharp short tick
Secondary weapon fire	Heavier thump
Special weapon fire	Long resonant build
Damage taken (light)	Single soft pulse
Damage taken (heavy)	Multiple sharp pulses
Shield raise	Building hum
Shield down	Descending pulse
Transporter Beam activation	Shimmer pattern (high-frequency rapid pulses)
Quantum Torpedo countdown tick	Escalating pulses
Quantum Singularity Event	Deep continuous rumble
Destruction	Sustained intense pattern
Power-up collect	Pleasant chirp
Victory	Triumphant cascade
Defeat	Single low thud

User can control haptic intensity (off / low / medium / high) in Settings.

14. Visual & Animation Design

Per-Ship Animations Required

- **Idle** — gentle bob / engine glow pulse
- **Thrusting** — engine streak / plasma jet trail
- **Turning** — slight thruster flare on opposite side
- **Primary weapon firing** — muzzle flash + projectile
- **Secondary weapon firing** — heavier flash + projectile
- **Special weapon firing** — unique per ship
- **Taking damage** — flash white briefly, shake
- **Low health** — periodic sparking / smoking trail
- **Shield active** — translucent hemisphere around ship, pulsing
- **Shield raising** — sweep animation
- **Shield lowering** — fade animation
- **Cloaked** — translucent ship outline (player view) or invisible (AI view)
- **Destruction** — multi-stage explosion (shockwave + debris + flash)

Arena Visual Elements

- **Parallax starfield** — multi-layer with depth
- **Planets** — 1–3 per arena, with gravity wells visible as subtle gradient rings
- **Planet rings** (Saturn-style on some)
- **Quantum Singularity debris** — fragmented hull pieces with glowing edges
- **Quantum Singularity warp effect** — chromatic aberration + lens distortion shader
- **Power-up icons** — color-coded translucent glow

HUD Visual Design

- **Mortal Kombat 1 style** life bars at top corners
- Ship name in retro-futuristic font (Orbitron or similar)
- Life / Shield / Battery bars stacked vertically
- Power-up icons below as translucent colored chips
- Match timer at top center
- Match score (e.g., "MATCH 1 OF 3 — WINS: 1–0") below timer
- "STAR MELEE" logo small at top center

DECISION POINT: For v1.0, ships will be rendered as **stylized polygon shapes** drawn programmatically (similar to the HTML mockup). This is recommended for MVP. Full sprite art / 3D models in v1.1+.

15. AI Opponent Design

AI Difficulty Levels

- **Cadet** (easy) — slow reaction, poor aim, rarely uses special weapons
- **Captain** (medium) — default — reasonable aim, occasional special usage
- **Admiral** (hard) — fast reaction, good aim, smart special usage, uses cover
- **Legendary** (very hard) — frame-perfect aim, predictive movement, optimal weapon usage

AI Behavior Tree

1. **Threat assessment** — health %, distance to opponent, distance to planets, power-ups available
2. **Movement** — pursuit, evasion, planet-orbit slingshot, power-up grab
3. **Combat** — fire primary in range, fire secondary at slower targets, use special when advantageous
4. **Defense** — raise shields when low health, evade homing missiles, transport away from incoming Quantum Torpedoes
5. **Mistake injection** — AI deliberately misses occasionally on easier difficulties

16. Settings Menu

Sections

1. **Audio**
 - Master Volume (slider 0–100%)
 - Music Volume (slider 0–100%)
 - SFX Volume (slider 0–100%)
 - Haptic Intensity (Off / Low / Medium / High) — iOS only
2. **Controls**
 - View Control Scheme diagram (visual reference of button layout)
 - Adjust button opacity (slider)
 - Adjust button size (slider)
 - Left-handed mode (mirror layout)
3. **Gameplay**
 - AI Difficulty (Cadet / Captain / Admiral / Legendary)
 - Match Length (90s / 120s default / 180s / 300s)
4. **How to Play** — interactive tutorial
5. **Ship Compendium** — link to Compendium screen
6. **Language & Localization** — language picker (English at launch; plan Spanish, French, German, Japanese, Korean, Mandarin for v1.1+)
7. **Leaderboard** — local win-loss records per ship + global Game Center leaderboard
8. **About**
 - App version

- Credits
- Link to djEnterprises website
- Link to Privacy Policy
- Link to Support / Contact
- Acknowledgments

17. Leaderboard System

Local Storage

- Stored in `UserDefaults` (or Core Data for richer data).
- Persists across app launches and updates.

Tracked Per Ship

- Total matches played
- Wins / Losses
- Win % (color-coded: green > 60%, yellow 40–60%, red < 40%)
- Best win streak
- Total damage dealt
- Total damage taken
- Quantum Torpedo kills (FATALITY count)
- Self-Destruct wins

Global (Game Center) — Optional v1.1

- Total wins across all ships
- Win streak
- Quantum Torpedo kills
- Highest single-match score

Player Scoring Formula (proposed)

```
match_score = (base_win_bonus) +  
              (damage_dealt × 0.5) -  
              (damage_taken × 0.25) +  
              (weapon_accuracy_pct × 100) +  
              (special_weapon_kills × 50) +  
              (time_remaining_bonus) +  
              (fatality_bonus × 200)
```


DECISION POINT: Confirm scoring formula. Recommended to make this tunable in a config file so it can be balanced post-launch.

18. In-App Purchases (StoreKit 2)

Initial v1.0 SKUs

- **Dominion Vanguard Pack** (\$2.99) — unlocks Void Reaper, Scarab Striker, Obsidian Maw
- **Shadow Fleet Pack** (\$2.99) — unlocks Wraith Phantom, Bone Spear, Crimson Tyrant
- **Bundle: All Ships** (\$4.99) — unlocks both packs (save \$0.99)
- **Weapon Enhancement: Primary +20%** (\$0.99 per ship) — boosts primary weapon damage
- **Weapon Enhancement: Shield +25%** (\$0.99 per ship) — boosts max shield
- **Weapon Enhancement: Battery +25%** (\$0.99 per ship) — boosts battery capacity
- **Cosmetic: Skin Pack** (\$0.99 per ship) — color/texture variations

StoreKit Implementation

- Use StoreKit 2 (async/await API).
- Server-side receipt validation NOT required for v1.0 (single-device purchases).
- Cloud sync via iCloud (CloudKit) so purchases persist across devices.
- Restore Purchases button in Settings.

19. Build Phases & Timeline

Phase 1: Core Combat Prototype (3–5 weeks)

- Xcode project setup, Git repo, SwiftUI menu shell
- SpriteKit scene with arena background
- Single ship with inertia movement
- Planets with gravity (and visual gravity wells)
- Touch controls (D-pad + 6 buttons, semi-transparent)
- Single weapon firing
- Basic damage and death
- **Milestone:** Playable 1v1 with a placeholder enemy that just exists.

Phase 2: Combat Depth (3–4 weeks)

- All 12 ships' geometric designs and stat profiles
- All primary, secondary, special weapons
- Shield, battery, life systems with HUD
- Mortal Kombat-style HUD
- Match timer + 2-of-3 series structure
- Power-up spawning and collection
- Pre-match 10-second free-movement countdown
- Pause functionality
- Game Over and victory screens with FATALITY logic
- AI opponent (Captain difficulty)
- **Milestone:** Full single-player melee experience.

Phase 3: Special Mechanics (2–3 weeks)

- Transporter Beam + Quantum Torpedo mechanic (with all defense behaviors)
- Quantum Singularity Event arena modification
- Cloaking Device + AI vs player visibility logic
- Self-Destruct with 4-second countdown
- All ship special weapons implemented
- Haptic feedback for all events
- **Milestone:** Every ship's full unique kit is playable.

Phase 4: Audio + Compendium + Polish (2–3 weeks)

- Music system (menu + match + special tracks)
- Per-ship sound effects (synthesized placeholders for v1.0)
- Ship Compendium with 3D rotatable models (low-poly SceneKit)
- Settings menu (audio sliders, controls, gameplay)
- Localization framework (English only at launch)
- Leaderboard local storage
- AI difficulty levels (Cadet / Admiral / Legendary)
- **Milestone:** Feature-complete v1.0 candidate.

Phase 5: IAP + macOS + Submission (2–3 weeks)

- StoreKit 2 integration (all ship packs and enhancements)
- Mac Catalyst build with keyboard controls
- App Store assets (icons, screenshots, preview video)
- TestFlight beta with 10–20 testers
- Bug fixes and polish
- App Store submission
- **Milestone:** Launched.

Total estimated build time: 12–18 weeks of focused part-time work.

20. File Structure (Proposed)

```

StarMelee/
■■■■ README.md
■■■■ STAR_MELEE_PLAN.md           ← this document
■■■■ star-melee-mockup.html       ← browser prototype reference
■■■■ .gitignore
■■■■ StarMelee.xcodeproj/
■■■■ StarMelee/
■   ■■■■ App/
■   ■   ■■■■ StarMeleeApp.swift
■   ■   ■■■■ AppDelegate.swift
■   ■   ■■■■ Info.plist
■   ■■■■ Scenes/
■   ■   ■■■■ MainMenuView.swift
■   ■   ■■■■ ShipSelectView.swift
■   ■   ■■■■ CombatScene.swift    ← SKScene
■   ■   ■■■■ PauseView.swift
■   ■   ■■■■ VictoryView.swift
■   ■   ■■■■ CompendiumView.swift
■   ■   ■■■■ SettingsView.swift
■   ■■■■ Gameplay/
■   ■   ■■■■ Ship.swift           ← entity
■   ■   ■■■■ Weapon.swift
■   ■   ■■■■ Projectile.swift
■   ■   ■■■■ Planet.swift
■   ■   ■■■■ PowerUp.swift
■   ■   ■■■■ QuantumTorpedo.swift
■   ■   ■■■■ QuantumSingularity.swift
■   ■   ■■■■ PhysicsEngine.swift
■   ■   ■■■■ AIController.swift
■   ■   ■■■■ MatchManager.swift  ← series logic (2-of-3)
■   ■■■■ Systems/
■   ■   ■■■■ InputSystem.swift    ← touch + keyboard
■   ■   ■■■■ HapticsSystem.swift
■   ■   ■■■■ AudioSystem.swift
■   ■   ■■■■ ScoringSystem.swift
■   ■   ■■■■ LeaderboardStore.swift
■   ■   ■■■■ IAPManager.swift     ← StoreKit 2
■   ■   ■■■■ LocalizationManager.swift
■   ■■■■ UI/
■   ■   ■■■■ HUDView.swift        ← Mortal Kombat-style HUD
■   ■   ■■■■ DPadView.swift
■   ■   ■■■■ ButtonClusterView.swift ← A/B/C/X/Y/Z
■   ■   ■■■■ LifeBarView.swift
■   ■   ■■■■ ShieldBarView.swift
■   ■   ■■■■ BatteryBarView.swift
■   ■   ■■■■ PowerUpIndicatorView.swift
■   ■■■■ Resources/
■   ■   ■■■■ Ships.json           ← ship stats config
■   ■   ■■■■ Weapons.json        ← weapon configs
■   ■   ■■■■ PowerUps.json
■   ■   ■■■■ Audio/

```

```

■ ■ ■ ■ Music/
■ ■ ■ ■ Localization/
■ ■ ■ ■ Models/
■ ■ ■ ■ ShipDefinition.swift
■ ■ ■ ■ WeaponDefinition.swift
■ ■ ■ ■ PowerUpDefinition.swift
■ ■ ■ ■ MatchState.swift
■ ■ ■ ■ StarMeleeTests/
■ ■ ■ ■ DamageCalculationTests.swift
■ ■ ■ ■ PhysicsTests.swift
■ ■ ■ ■ MatchStateTests.swift
■ ■ ■ ■ StarMeleeUITests/

```

21. Reference Repositories

These are GPL-licensed and **for study only** — do not copy substantial code blocks. Reimplement mechanics from scratch in Swift.

Primary Study Reference: The Ur-Quan Masters (UQM)

- **Repo:** <https://github.com/JHGuitarFreak/UQM-MegaMod>
- **What to study:** Ship physics (thrust, inertia, turn), weapon spawn/lifecycle, AI behavior tree, projectile collision, planet gravity computation, special-weapon implementations.
- **License:** GPL v2 — do not import code directly into Star Melee.

Secondary Reference: TW-Light

- **Repo:** <https://github.com/Yurand/tw-light>
- **What to study:** Pure melee loop, camera handling, multi-ship combat structure.

22. Workflow with Claude Code

Recommended Session Pattern

1. Read `STAR_MELEE_PLAN.md` at the start of every session.
2. Identify the current Phase and uncompleted milestone.
3. Build in vertical slices — one full feature end-to-end before moving on.
4. Commit after every working feature with descriptive Git messages.
5. Run on a physical device frequently — touch feel cannot be evaluated in Simulator.
6. Update this document when scope changes or decisions are made.

Sample Claude Code Prompts (for the user to copy/paste)

Phase 1 kickoff:

"Read STAR_MELEE_PLAN.md and star-melee-mockup.html. Then scaffold a new Xcode project for the iOS app Star Melee using Swift + SpriteKit + SwiftUI. Set up the basic SwiftUI main menu (per Section 4 and 16 of the plan), and create an empty CombatScene SKScene with a placeholder arena background. No ships yet. Commit when complete."

Phase 1 mid-build:

"Implement the Ship class per Section 5 of STAR_MELEE_PLAN.md. Add inertia physics matching the behavior of the HTML mockup. Create one playable ship (Aegis Cruiser) with the touch D-pad controls per Section 9. Do NOT implement weapons yet. Commit when ship can move."

Phase 2 milestone — ship select:

"Build the Ship Select screen per Section 4 + Section 19 (Phase 2). Read Ships.json for the ship roster. Display all 12 ships in a grid with their stat profiles. Selecting a ship and tapping CONFIRM should launch the CombatScene with the chosen ship."

23. Critical Fixes from HTML Mockup Feedback

The user identified that in the previous mockup, the player dies almost immediately, and the countdown number blocked visibility. Address all of these in implementation:

1. **Starting ship positions** must be in safe zones — at least 200px from any planet, one ship per side of arena.
2. **Pre-match 10-second practice countdown** with the exact rules from Section 4:
 - Player and AI can move and fire **primary (A)** and **secondary (B)** weapons.
 - AI fires at reduced frequency during practice (~60% primary rolls, ~20% secondary rolls).
 - **Locked during countdown:** C (special), Z (speed boost), A+B (Transporter), B+C (Cloak), A+C (Self-Destruct), and power-up spawning.
3. **Gravity ramping** — planet gravitational pull starts at 0 and ramps to full strength over the last 5 seconds of the countdown. Players should not get pulled into planets before they understand the controls.
4. **Visible "PRACTICE" banner** during countdown with hint text explaining what works (e.g., "Move and fire A (primary) / B (secondary) to test your ship. Specials, Boost, Transporter, Cloak unlock at MATCH START.").
5. **Semi-transparent countdown digit** — render the large 10/9/8/... number at ~35% opacity with a thin outline ring so the arena and ships remain visible behind it. Do not fully obscure the screen.
6. **Initial ship orientation** facing away from the nearest planet.
7. **Tutorial / How to Play** screen accessible from first launch and from the pause menu.
8. **Implementation pattern (for Claude Code):** in the ship update function, pass a single `allowSpecials: Bool` flag. Primary and secondary firing should never check this flag — they always work in any playing phase. Specials, combos, and boost must check it.

24. App Store Submission Checklist

Before submitting v1.0:

- ☐ All 12 ships playable and balanced
- ☐ All weapon systems functional
- ☐ All power-ups spawning and collectible
- ☐ Transporter Beam + Quantum Torpedo + Singularity Event working
- ☐ FATALITY trigger working
- ☐ 2-of-3 match series logic verified
- ☐ Touch controls fully functional on iPhone and iPad
- ☐ Keyboard controls fully functional on macOS
- ☐ All Settings options working
- ☐ StoreKit 2 IAP tested on sandbox account
- ☐ Restore Purchases working
- ☐ Privacy Policy URL live
- ☐ Support email + URL live
- ☐ App Store description, keywords, screenshots, preview video done
- ☐ App icon designed (1024×1024)
- ☐ Privacy Manifest filled out
- ☐ No copyrighted content (audit pass)
- ☐ TestFlight beta tested by 10+ external users
- ☐ Performance verified on iPhone 12, iPad Air 4, Apple Silicon Mac
- ☐ Crash-free rate > 99.5% in TestFlight

25. Open Decision Points Summary

Collected from throughout this document. Resolve before or during the relevant Phase:

1. **Ship distribution: free vs. paid** (Section 1, 5) — Recommend 6 free + 6 paid split.
2. **Swift + SpriteKit confirmation** (Section 2) — Default Yes.
3. **Universal Transporter Beam or IAP-gated** (Section 5, 6) — Recommend universal with battery cost.
4. **Damage formula final values** (Section 6) — Build as JSON config; tune in playtesting.
5. **3D model fidelity in Compendium** (Section 11, 14) — Recommend low-poly SceneKit for MVP.
6. **Crimson Tyrant being a "premium" ship** (Section 5) — Decision needed: pay-to-win perception risk.
7. **Scoring formula final values** (Section 17) — Build as JSON config; tune in playtesting.
8. **Music asset strategy** (Section 12) — Royalty-free placeholder for v1.0; commission composer for v1.1.
9. **Localization timing** (Section 16) — English only at launch is fine.

10. **Game Center leaderboards in v1.0** (Section 17) — Recommend defer to v1.1.

26. Closing Notes

This document is the complete blueprint for Star Melee v1.0. It is designed to be read end-to-end by Claude Code, and to be the source of truth for every decision in the build.

For Claude Code: When in doubt about an unspecified behavior, default to the behavior closest to Star Control II's Super Melee mode. If still ambiguous, stop and prompt the user.

For Dan: Read the open decision points in Section 25. Resolve the must-decide ones (free vs. paid split, Crimson Tyrant tier) before Phase 2. Everything else can be decided as it comes up.

This game has potential. The combat depth, freemium model, and Mortal Kombat-style presentation are a strong combination. Build it tight, ship the MVP, then iterate based on player data.

Document compiled May 23, 2026. All sections current as of this date.

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